

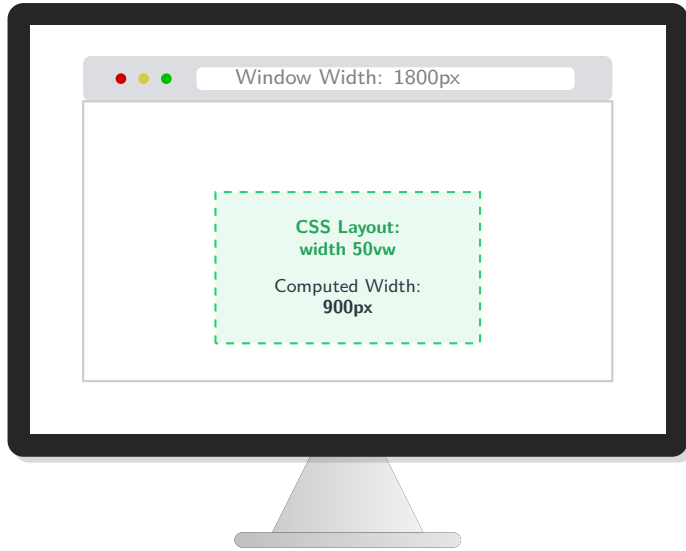
Fluid Layout Independence of Density Descriptors

Why CSS width: 50vw changes nothing about the engine's 2x source selection.

Two Different Viewports, Same Hardware Density

 **User A: 4K Retina Display**

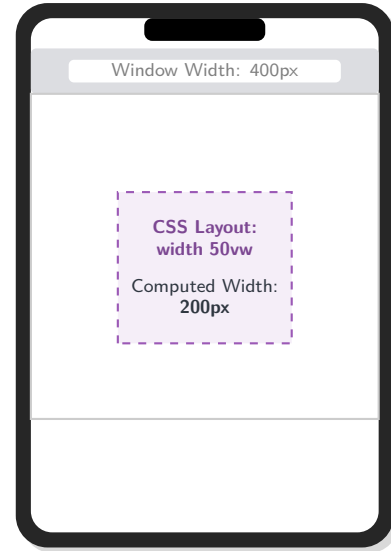
Hardware DPR = 2.0



Downloads: image-2x.jpg

 **User B: High-End Mobile**

Hardware DPR = 2.0



Downloads: image-2x.jpg

The Underlying Architecture Pipeline

1. HTML Parsing

Engine encounters `<img>`
sees `srcset="... 2x"`

2. Network Dispatch

Engine queries OS DPR.
DPR = 2.0. Engine **immediately**
requests `image-2x.jpg`.

3. CSS Computation

DOM calculates fluid layout.
width: 50vw is applied.
(Too late to change image!)

Pre-load Scanner Phase (Fast)

Rendering Phase (Slow)